

Partial Equilibrium With a Competitive Market

Partial Equilibrium: Concerned with one or a subset of markets. Supply and demand Analysis.

$Q_d = D(p, \alpha)$; α : a bunch of variables that affect demand. (price of other ^{related} goods, population, time of year, etc.)

$Q_s = S(p, \beta)$ β : a bunch of variables that affect supply (input prices, environmental condition, regulation etc.)

α and β are not mutually exclusive.

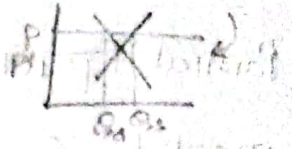
In equilibrium, $Q_d = Q_s$

Sometimes we call the price when the markets are "in equilibrium" — the "equilibrium price" or "the market clearing price" — let's call this price p^* .

If $p < p^*$, $Q_d > Q_s \rightarrow$ we have a shortage

If $p > p^*$, $Q_s > Q_d \rightarrow$ we have a surplus.

Partial equilibrium takes longer than the instant disequilibrium solution.



General equilibrium is slower than partial.

Shortages and surpluses are caused by prices not adjusting fast enough.

from one eq^m to the next, the condition to restore

the eq^m is $dQ_d = dQ_s$

Example: Price of Canned beverage when the price of aluminium increases. $\uparrow$ — this affects β

$$Q_d = D(p, \alpha) \quad Q_s = S(p, \beta)$$

Price of aluminium is not in α , but is in β .

$$d\alpha = 0$$

$$d\beta > 0 \quad (\text{since price increase})$$

P_a = price of aluminium

$$dQ_d = \frac{\partial D}{\partial P} \cdot dP + \frac{\partial D}{\partial \alpha} \cdot d\alpha$$

$$\Rightarrow dQ_d = \frac{\partial D}{\partial P} \cdot dP$$

$$dQ_s = \frac{\partial S}{\partial P} \cdot dP + \frac{\partial S}{\partial \beta} d\beta$$

$$dQ_s = \frac{\partial S}{\partial P} \cdot dP + \frac{\partial S}{\partial P_a} dP_a \quad [\text{Here, } \beta = P_a]$$

Restore eqn: $dQ_d = dQ_s$

$$\Rightarrow \frac{\partial D}{\partial P} \cdot dP = \frac{\partial S}{\partial P} \cdot dP + \frac{\partial S}{\partial P_a} dP_a$$

we are interested in $\frac{dP}{dP_a}$.

$$\Rightarrow \left[\frac{\partial D}{\partial P} - \frac{\partial S}{\partial P} \right] dP = \frac{\partial S}{\partial P_a} \cdot dP_a$$

$$\Rightarrow \frac{dP}{dP_a} = \frac{\partial S / \partial P_a}{\left[\frac{\partial D}{\partial P} - \frac{\partial S}{\partial P} \right]}$$

Correct, but in the terms of slopes, we'd like elasticities

instead.

$$\frac{dP}{dP_a} \cdot \frac{P_a}{P} = \frac{\frac{\partial S}{\partial P_a} \cdot \frac{P_a}{P} \cdot \frac{P}{Q}}{\frac{\partial D}{\partial P} \cdot \frac{P}{Q} - \frac{\partial S}{\partial P} \cdot \frac{P}{Q}}$$

$$\Rightarrow E_{P, P_a} = \frac{E_{Q, P_a}^S}{E_{Q, P}^D - E_{Q, P}^S}$$

$$\Rightarrow E_{P, P_a} = \frac{-E_{Q, P_a}^S}{E_{Q, P}^S - E_{Q, P}^D}$$

$$= \frac{-E_{Q, P_a}^S}{E_{Q, P}^S} + \frac{E_{Q, P_a}^S}{E_{Q, P}^D} \quad [\because P_a \uparrow \Rightarrow S \downarrow]$$

tax impose on production.

the same conclusion

Incidence of taxation

Once a sales tax is involved, the price the buyers pay is different than the price suppliers keep.

2 different types of tax

— Proportional $\tau^{(tax)}$ $P_b = (1 + \tau) P_s$

— Unit tax t $P_d = P_s + t$

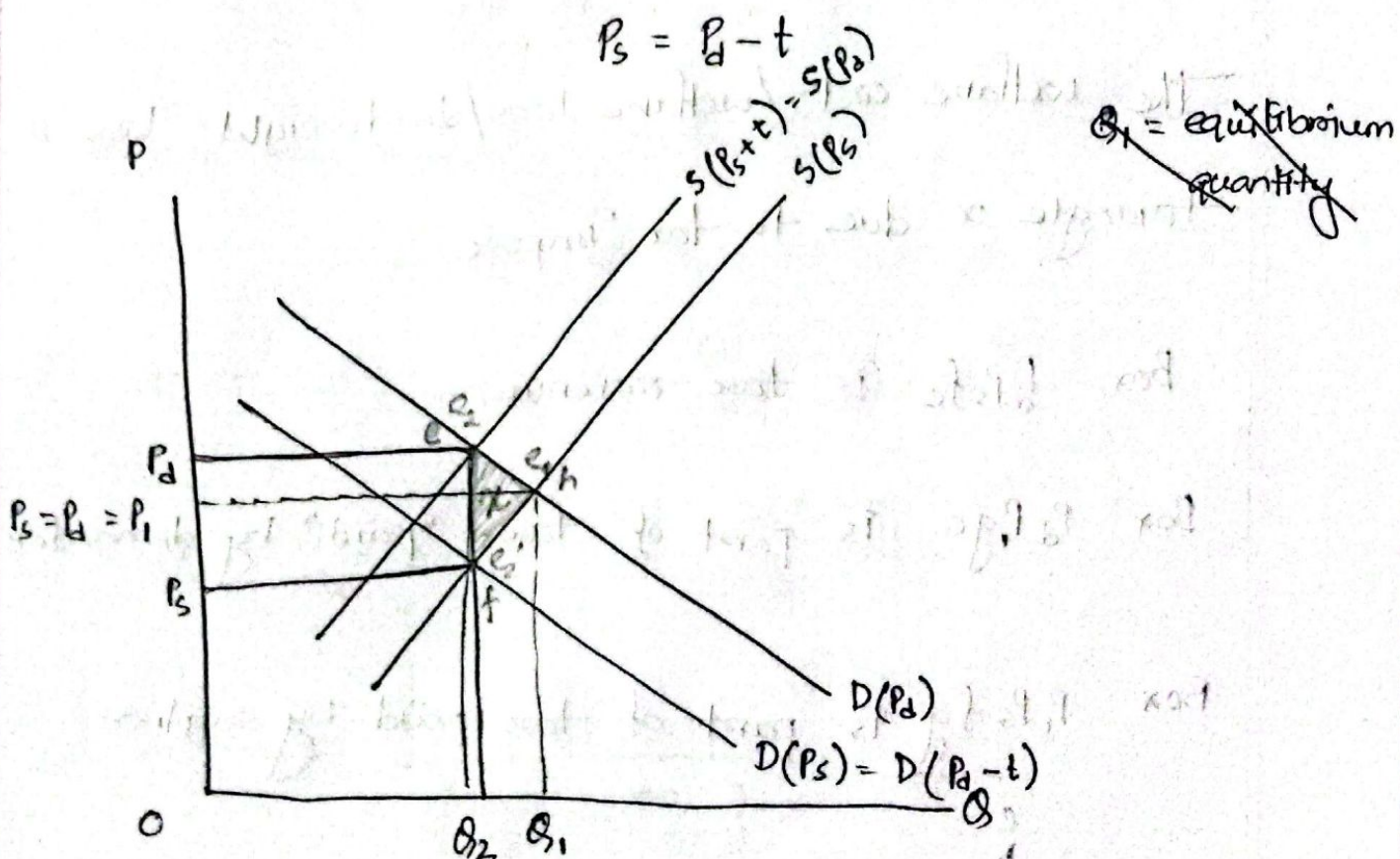


Fig: unit tax graphically

$Q_1 =$ No tax equilibrium quantity

If $P_D \neq P_S$ assume a unit tax of t

In the sense of relative price, ~~as~~ now with higher relative price, the demand shift ~~is~~ leftward.

But this is not a stable solution.

$Q_2 =$ eq^m quantity with tax t .

The welfare cost / welfare loss / deadweight loss is the triangle x due to tax impose.

Box $P_d P_s f e$ is tax revenue

Box $P_d P_i g e$ is part of tax "paid" by demanders

Box $P_i P_s f g$ is part of tax paid by suppliers



deadweight welfare loss

welfare cost of the tax

Tax Incidence: Partial Equilibrium, Competitive Model

$$Q_d = D(P_d, \alpha) \quad Q_s = S(P_s, \beta)$$

$$P_d = P_s + t$$

$$dP_d = dP_s + dt \quad \text{or} \quad dP_s = dP_d - dt$$

$$dQ_d = \frac{\partial D}{\partial P} \cdot dP_d + \frac{\partial D}{\partial \alpha} \cdot d\alpha$$

$$dQ_s = \frac{\partial S}{\partial P} \cdot dP_s + \frac{\partial S}{\partial \beta} \cdot d\beta$$

To restore equilibrium,

$$dQ_d = dQ_s$$

$$\frac{\partial D}{\partial P} \cdot dP_d = \frac{\partial S}{\partial P} \cdot dP_s = \frac{\partial S}{\partial P} [dP_d - dt]$$

$$\Rightarrow dP_d \left[\frac{\partial S}{\partial P} - \frac{\partial D}{\partial P} \right] = \frac{\partial S}{\partial P} \cdot dt$$

$$\Rightarrow \frac{dP_d}{dt} = \frac{\frac{\partial S}{\partial P}}{\left[\frac{\partial S}{\partial P} - \frac{\partial D}{\partial P} \right]}$$

$$= \frac{dP_d}{dt} = \frac{\frac{\partial S}{\partial P} \cdot \frac{P}{Q}}{\left[\frac{\partial S}{\partial P} - \frac{\partial D}{\partial P} \right] \cdot \frac{P}{Q}}$$

$$\rightarrow \frac{dP_d}{dt} = \frac{E_{q,p}^S}{E_{q,p}^S - E_{q,p}^D}$$

Similarly

$$\frac{dP_s}{dt} = \frac{E_{q,p}^D}{E_{q,p}^S - E_{q,p}^D}$$

☐ Tobacco Problem:

Raise the tax per pack by \$1

$dt = 1$, suppose $E^D = -0.2$

E^S is elastic for easy math. $E^S = 1.8$

$$\frac{dP_d}{dt} = \frac{1.8}{1.8 + 0.2} = 0.9$$

The Production function, $Q = (KL)^{0.25}$ has corresponding long-run cost function, $C(q, w) = 2q^{\frac{1}{3}} \sqrt{w}$.

Assume that the firm is price taker in inputs and outputs. Find the firm's supply curve.

$$C(q, w) = 2q^{\frac{1}{3}} \sqrt{w}$$

$$\text{Marginal cost } MC = 4q^{\frac{1}{3}} \sqrt{w}$$

$$\text{In price taking firm, } P = MC$$

$$\Rightarrow P = 4q^{\frac{1}{3}} \sqrt{w}$$

$$\Rightarrow q = \frac{P}{4\sqrt{w}} \quad ; \text{ Firm's supply curve.}$$

Show that if the production function is homothetic, marginal cost increases with an increase in all input prices.

Pre condition $\rightarrow$ Homogeneous function

A competitive industry consists of 2000 identical firms each with short run total cost by $C_{SR}(q) = 0.5q^2 + 2q + 90$.

a. What is the short run supply curve for each firm?

b. What is the industry's short-run supply?

Suppose market demand

$$Q_d = 64000 - 1000p$$

c. Find the equilibrium price and quantity.

d. What is each firm's short run profit?

a. $C_{SR}(q) = 0.5q^2 + 2q + 90$

$$MC = q + 2$$

for a competitive firm,

$$P = MC$$

$$\Rightarrow P = q + 2$$

$$\Rightarrow q = P - 2 \quad ; \quad \text{firm's short run supply.}$$

b) Here,

$n = \text{number of firms in the industry} = 2000$

$$\underline{Q_s = nq = 2000(p-2) = 2000p - 4000.}$$

c) In the equilibrium,

$$\underline{Q_d = Q_s}$$

$$\Rightarrow 64000 - 1000p = 2000p - 4000$$

$$\Rightarrow 3000p = 68000$$

$$\Rightarrow p = \frac{68000}{3000}$$

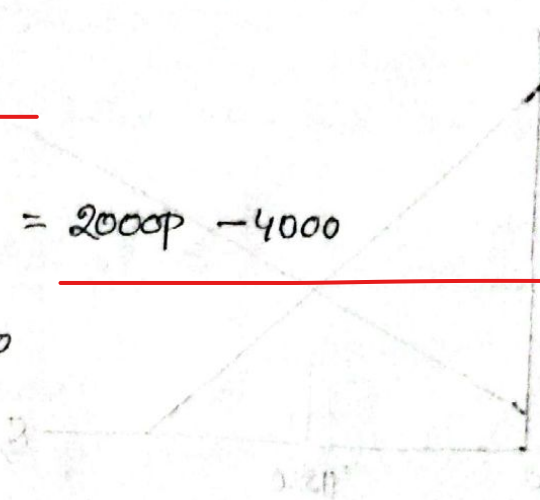
$$\Rightarrow \underline{p^* = 22.67}$$

$$Q_d = 64000 - 1000(22.67)$$

$$= \underline{41330}$$

$$q = \frac{41330}{2000}$$

$$= \underline{20.66}$$

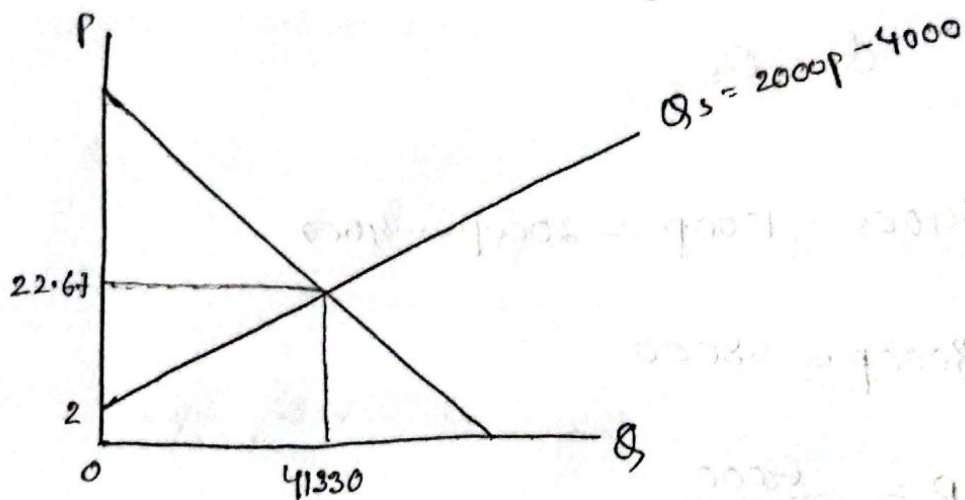


$$d \text{ profit} = pq - c(q)$$

$$= 22.67 \times 20.66 - [0.5(20.66)^2 + 2(20.66) + 20]$$

$$= 476.07 - 352.5 = 123.57$$

e. What is the short-run industry-wide producer surplus?



Producer surplus,

$$PS = \frac{1}{2} \times 41330 \times (22.67 - 2)$$

$$= 427176.55$$

2. b) $q = 2K + L$

$$\sigma = \infty$$

$$MRTS = \frac{MP_L}{MP_K} = \frac{1}{2} \quad \text{and} \quad \Delta MRTS = 0$$

$$\sigma = \frac{\frac{1}{2}}{\frac{1}{2}} \cdot \frac{MRTS}{\Delta MRTS} = \frac{\frac{1}{2}}{0} = \infty$$

b) $C_{LR} = \min(wq, v \cdot \frac{q}{2})$

c)

$$q = \frac{4K \cdot K \sqrt{\frac{v}{w}}}{K \sqrt{\frac{v}{w}} + K}$$

$$q = \frac{4K^* \sqrt{\frac{v}{w}}}{K(\sqrt{\frac{v}{w}} + 1)}$$

$$\Rightarrow K^* = \frac{q(\sqrt{\frac{v}{w}} + 1)}{4\sqrt{\frac{v}{w}}}$$

$$C_{LR} = wL + vK$$

$$= w \left[\frac{q(1 + \sqrt{\frac{v}{w}})}{4\sqrt{\frac{v}{w}}} \right] + v \left[\frac{q(1 + \sqrt{\frac{v}{w}})}{4\sqrt{\frac{v}{w}}} \right]$$

$$q = \frac{4L \cdot L \sqrt{\frac{v}{w}}}{L(\sqrt{\frac{v}{w}} + 1)}$$

$$q = \frac{4L(\sqrt{\frac{v}{w}})}{(\sqrt{\frac{v}{w}} + 1)}$$

$$\Rightarrow L^* = \frac{q(\sqrt{\frac{v}{w}} + 1)}{4\sqrt{\frac{v}{w}}}$$

$$= q \left(1 + \sqrt{\frac{w}{v}}\right) \cdot \frac{\sqrt{wv}}{4} + q \left(1 + \sqrt{\frac{v}{w}}\right) \sqrt{\frac{wv}{4}}$$

$$= q \left(\frac{\sqrt{v} + \sqrt{w}}{\sqrt{v}}\right) \cdot \frac{\sqrt{wv}}{4} + q \left(\frac{\sqrt{w} + \sqrt{v}}{\sqrt{w}}\right) \frac{\sqrt{wv}}{4}$$

$$= q \sqrt{w} \cdot \left(\frac{\sqrt{v} + \sqrt{w}}{4}\right) + q \sqrt{v} \cdot \left(\frac{\sqrt{w} + \sqrt{v}}{4}\right)$$

$$= \frac{q}{4} (\sqrt{wv} + w + v + \sqrt{wv})$$

$$= \frac{q}{4} ((\sqrt{w})^2 + 2\sqrt{wv} + (\sqrt{v})^2)$$

$$= \frac{q}{4} (\sqrt{w} + \sqrt{v})^2$$

$$\# \quad \frac{\partial \mathcal{H}^i(q, w)}{\partial w_i} = \frac{\partial l_i^*}{\partial q} > 0 \quad \left[-\frac{\partial l}{\partial q} > 0 \right]$$

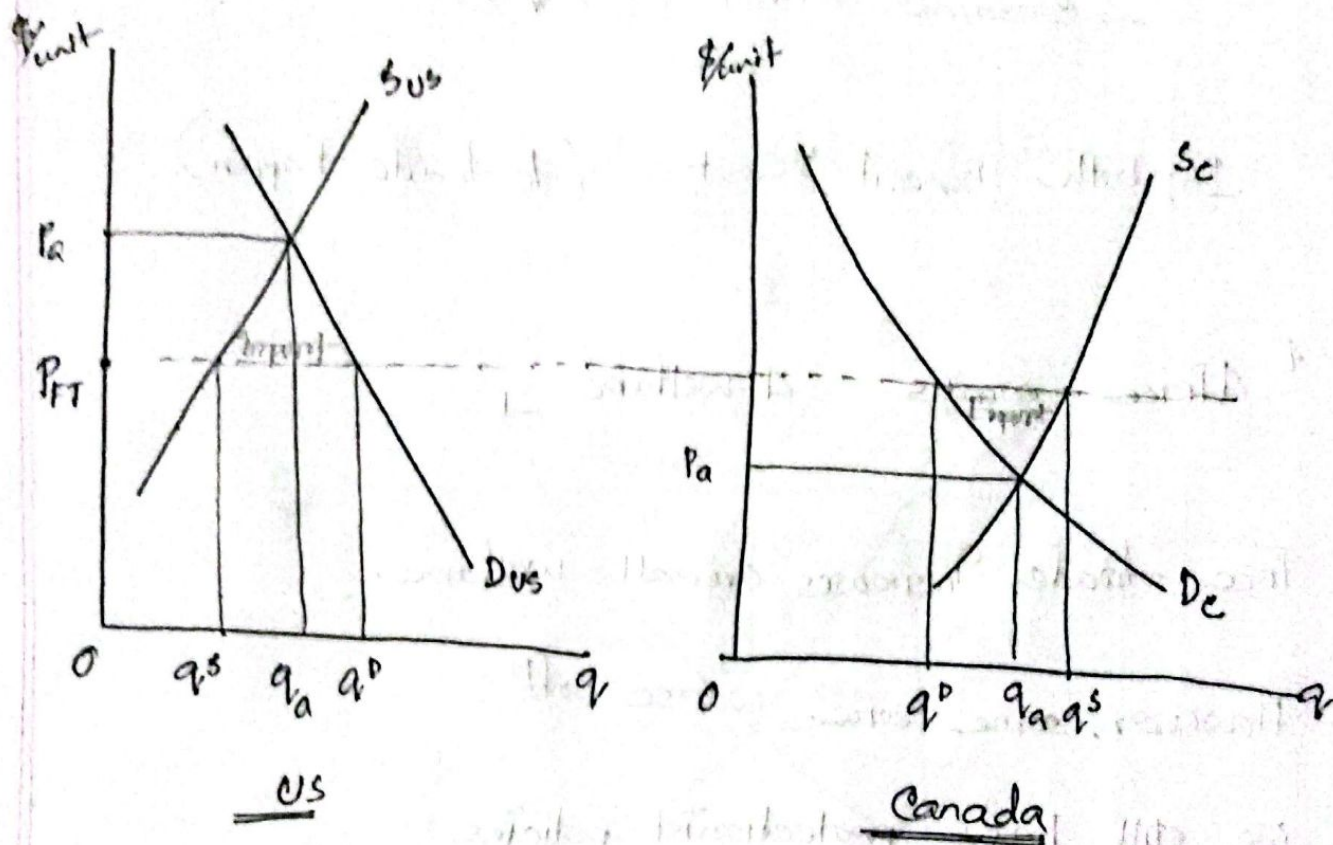
$$\frac{\partial \mathcal{H}^i(q, v)}{\partial v_i} = \frac{\partial k_i^*}{\partial q} > 0 \quad \left[-\frac{\partial k}{\partial q} > 0 \right]$$

* Young's theorem

* Envelope theorem

Applied Equilibrium (Partial) Analysis

International Trade : 2 country Model



a = Autarky — No trade

Free-trade (FT): Price adjustment to Imports = Exports

Import $\rightarrow$ US || Export $\rightarrow$ Canada

Autarky & Trade:

Importing Country:

— Consumer benefit $(CS) \uparrow$

— Producer Suffer $(PS) \downarrow$

Exporting country:

- Producer Benefit (P_s) $\uparrow$
- Consumer Suffer (C_s) $\downarrow$

In both, Benefit $>$ Cost (to trade happen)

Price means welfare

Free trade increases overall welfare.

However, some are worse off.

We still have protectionist policies.

Protectionist policy $\rightarrow$ Policy to reduce import.

Expansionary policy $\rightarrow$ Policy to increase exports.

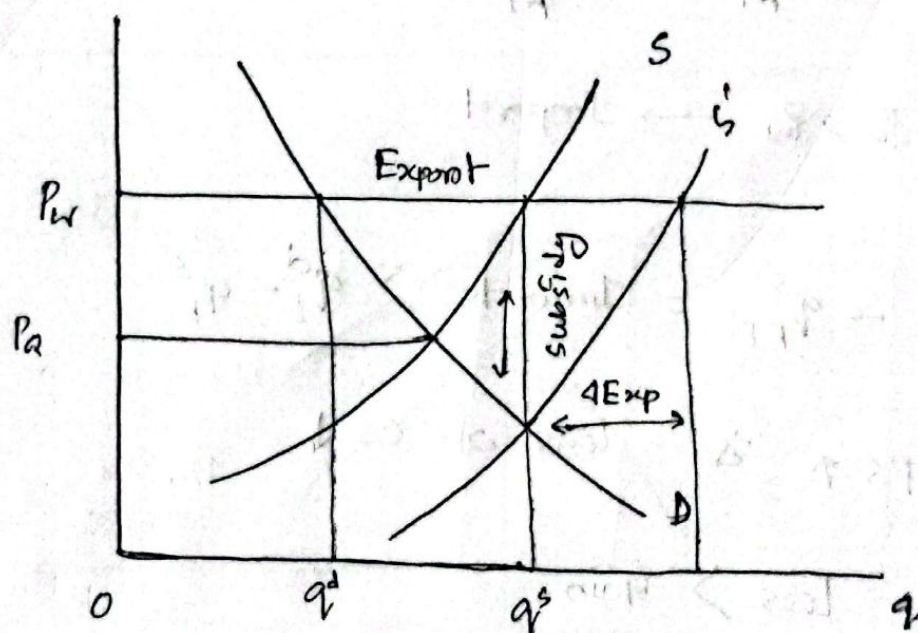
Tariff / Import duty, quota, voluntary import restraint.

Export Subsidy, currency devaluation, Indirect subsidies by governments.

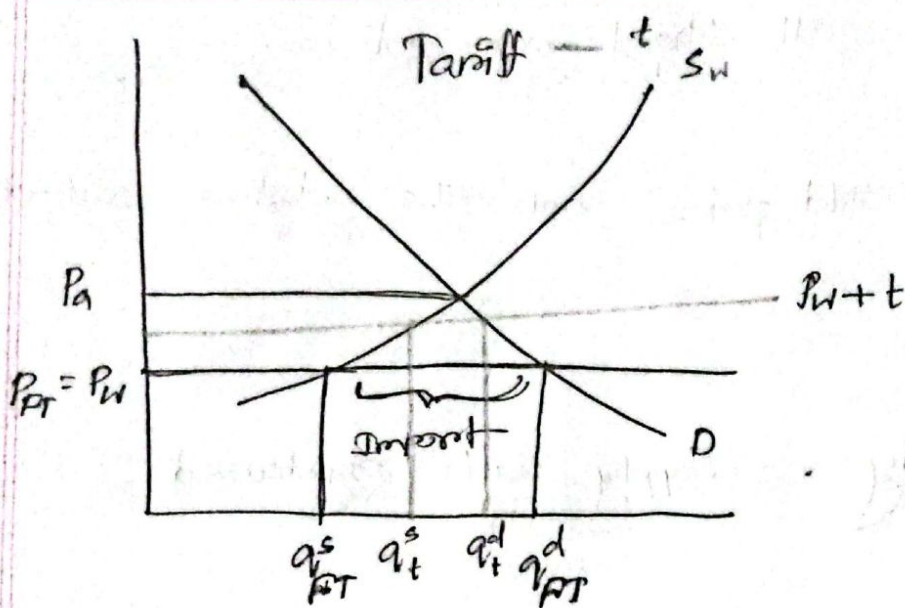
— these are all short-run policies.

[What about world price, not the relative / autarky price.]

Export subsidy — Supply shifts rightward.



If $P_W > P_A \rightarrow$ Export under free trade



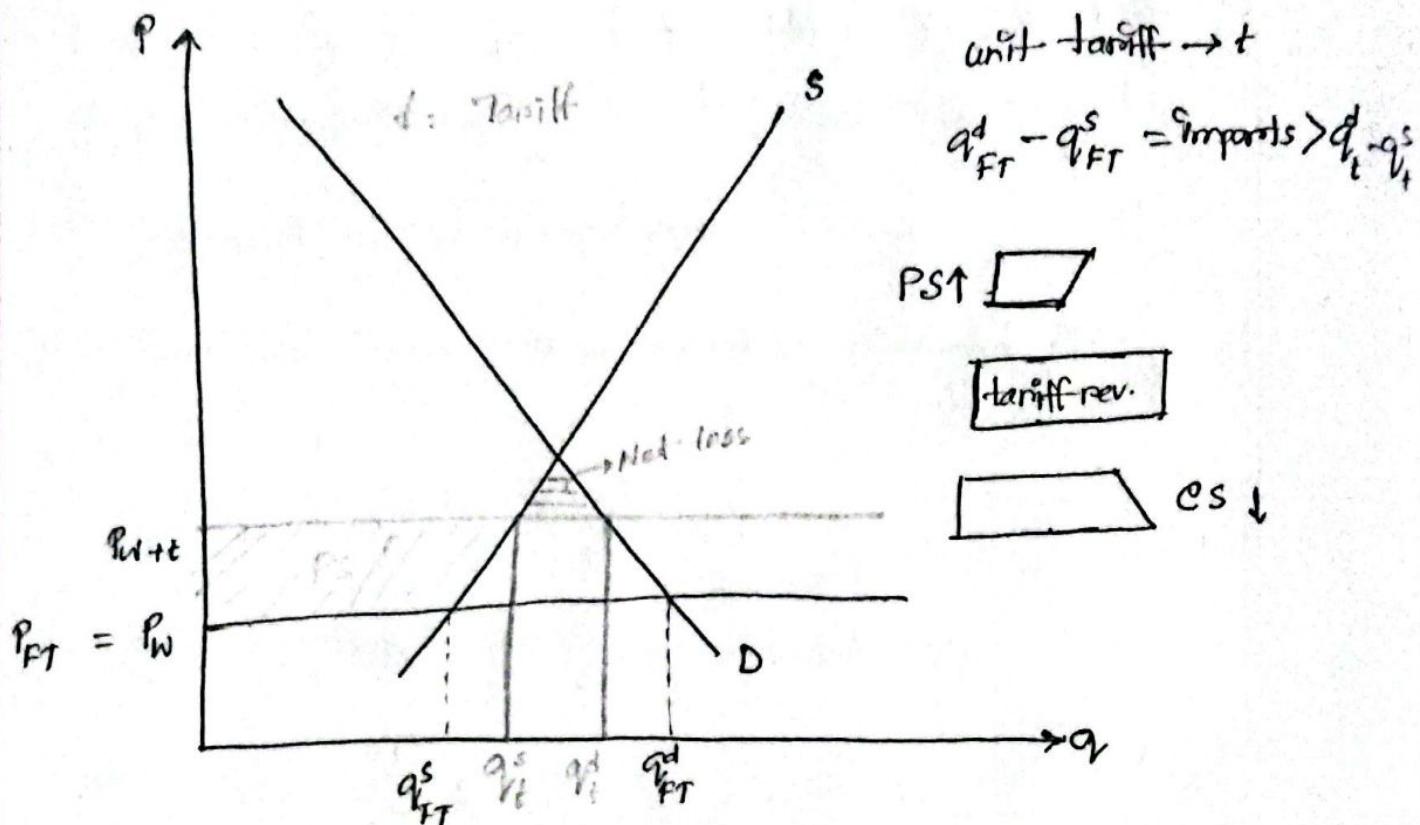
If $P_a > P_w \rightarrow \text{Import}$

$$q_{FT}^d - q_{FT}^s = \text{Imports} > q_t^d - q_t^s$$

$PS \uparrow$ & loss of $CS \downarrow$

loss > Gain

Advantageous for domestic producer.



Results from a tariff

- Domestic production in that industry increases (more domestic jobs in that industry)
- Higher producer surplus.

Possibly help with adopting a new technology.

- ESOP $\rightarrow$ Employment stock ownership plans

- Government Revenue

$\rightarrow$ Gets the attention of our trading partners.

- Imports decrease
- Higher domestic price / lower domestic consumption
- Loss of consumer surplus.
- Welfare loss:

If the good in production is an input in different domestic industry, you will have higher costs and decreased production (jobs) in those industries.

Often the case that more domestic jobs are lost in the related industries than are saved in the protected industry.

Quota's benefit goes to exporters/importers depends on the policy.

Quota combination of importers & exporters function.

cost of protection policy

Mathematical Model of Partial Equilibrium Trade Model

$$\underline{Q_d = 100 - P}$$

$$\underline{Q_s = \frac{P}{2} - 5}$$

$$\underline{P = 100 - Q_d}$$

$$\underline{P = 2Q_s + 10}$$

a. Find the autarky equilibrium.

$$100 - Q_d = 2Q_s + 10$$

$$Q_d = Q_s$$

$$\underline{3Q = 90}$$

$$\underline{Q_a = 30}$$

$$\underline{P_a = 70}$$

$$W_a = \frac{1}{2} (100 - 10) (30) = 1350$$

or

$$CS_a = \frac{1}{2} (100 - 70) (30) = 450$$

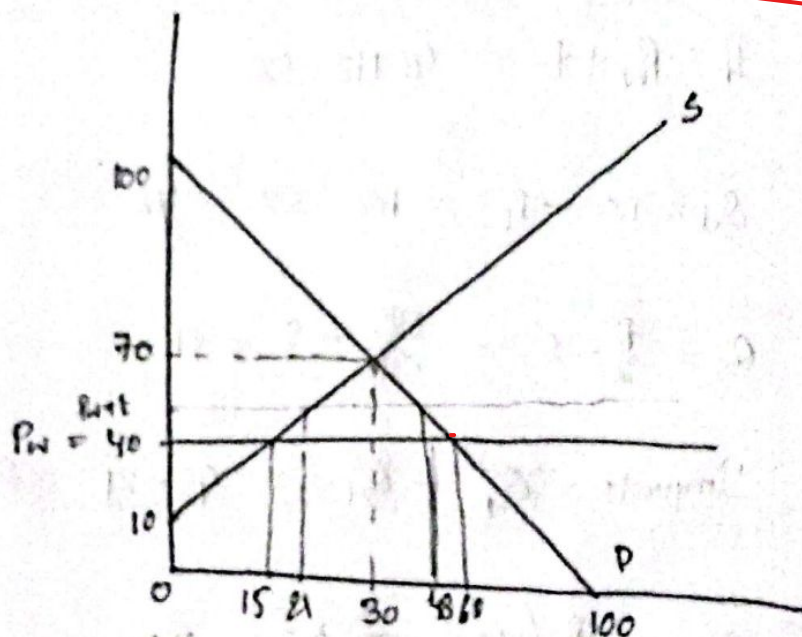
$$PS_a = \frac{1}{2} (70 - 10) (30) = 900$$

$$W_a = CS_a + PS_a = 450 + 900 = 1350$$

b. Small open Economy. $P_w = 40$.

Find the Q^d , Q^s , Imports, welfare under free trade.

$Q^d = 100 - 40$



$$\underline{Q_{FT}^d = 100 - 40 = 60}$$

$$\underline{Q_{FT}^s = \frac{40}{2} - 5 = 15}$$

$$\underline{\text{Imports} = Q_{FT}^d - Q_{FT}^s = 60 - 15 = 45}$$

$$\underline{W_{FT} = CS_{FT} + PS_{FT}}$$

$$= 1800 + 225$$

$$\underline{= 2025}$$

$$CS_{FT} = \frac{1}{2} (100 - 40) 60$$

$$= 1800$$

$$PS_{FT} = \frac{1}{2} (40 - 10) 15$$

$$= 225$$

c. Impose tariff of $12/\text{unit}$. What is tariff revenue?

$P_t = ?$ $Q_d = ?$ $Q_s = ?$ Imports = ? Welfare = ?

$$P_t = P_w + t = 40 + 12 = 52$$

$$Q_d = 100 - P_t = 100 - 52 = 48$$

$$Q_s = \frac{P}{2} - 5 = \frac{52}{2} - 5 = 21$$

$$\text{Imports} = Q_d^d - Q_s^s = 48 - 21 = 27$$

$$CS_t = \frac{1}{2} (100 - 52) 48 = 1152$$

$$PS_t = \frac{1}{2} (52 - 10) 21 = 441$$

Now,

$$\text{Tariff revenue} = 12 \times 27$$

$$\text{Welfare}_t = CS_t + PS_t + \text{Tariff revenue}$$

$$= 324$$

$$= 1152 + 441$$

$$W_t = 1152 + 441 + 324$$

$$= 1593$$

$$= 1917$$

$$W_t < W_{fr}$$

Welfare cost of tariff is $2025 - 1917 = 108$

d. Quota or VER (Voluntary Export Restraint) of 21.

① $\text{tariff revenue} = t \times \text{imports}$

$$= 12 \times 27 = 324$$

$$W_t = CS + PS + \text{tariff revenue}$$

$$= 1152 + 441 + 324 = 1917$$

Welfare cost of tariff is $2025 - 1917 = 108$

$$\text{or } \frac{1}{2}(12)(21-15) + \frac{1}{2}(12)(60-48) = 108.$$

② Imports = $Q_d - Q_s = 21$

$$100 - P - (P_2 - 5) = 21$$

$$\Rightarrow \frac{3}{2}P = 84$$

$$\Rightarrow P_Q = 56$$

$$Q_d = 100 - 56 = 44$$

$$Q_s = \frac{56}{2} - 5 = 23$$

$$\text{Value of right to import} = (56 - 40)(21) = 336$$

$$CS_B = \frac{1}{2} (100 - 56)(44) = 968$$

$$PS_B = \frac{1}{2} (56 - 10)(23) = 529$$

$$W_B = CS_B + PS_B = 968 + 529 =$$

Perfectly Competitive (Walrasian) Market

1. large numbers of relatively small buyers and sellers.
2. Homogeneous product.
3. Perfect information on price and quantity.
4. free entry and exit.
5. No transaction costs.

(1-3) everyone is a price taker $\rightarrow$ one market price that no one can influence.

4 - zero long run profit (economic) $\rightarrow$ only covering opportunity cost.

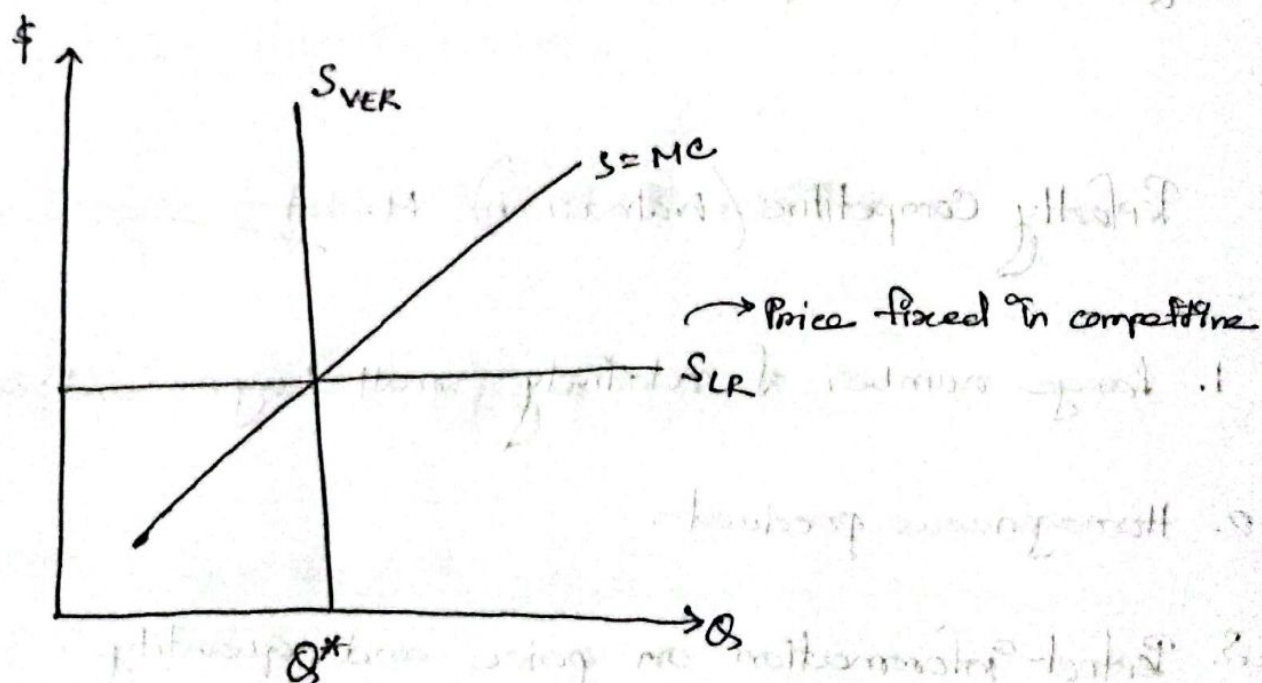
Market Dynamics:

very short-run $\rightarrow$ output is fixed

short-run $\rightarrow$ capacity (or number of firms) is fixed

Industry MC (just S)

long-run $\rightarrow$ capacity (or number of firms) can change



$$\pi_{SR} \geq 0$$

Algebraic Example:

1000 identical firms operating in a market, $n = 1000$

(With many more identical firms outside of the market, ready to enter) with costs $c(q, w) = q^2 + 4q + 16$

a. find the short-run supply.

b. Suppose demand is $Q_d = 18000 - 500p$

find the S.R. equilibrium.

c. Expect entry or exit?

d. find the long-run equilibrium.

a. $c(q, w) = q^2 + 4q + 16$

$$\Rightarrow MC = \frac{\partial c(q, w)}{\partial q} = 2q + 4$$

for optimization,

$$MR = MC$$

$$\Rightarrow P = 2q + 4 \quad [\because MR = P]$$

$$\Rightarrow 2q = P - 4 \Rightarrow q = \frac{P}{2} - 2 \quad ; \text{ individual firm's SR supply.}$$

$$Q_s = nq = 1000 \left(\frac{p}{2} - 2 \right)$$

$$\Rightarrow Q_s = 500p - 2000$$

b. In equilibrium,

$$Q_d = Q_s$$

$$\Rightarrow 18000 - 500p = 500p - 2000$$

$$\Rightarrow 1000p = 20000$$

$$\Rightarrow p^* = 20$$

$$\& \quad Q^* = 18000 - 500(20)$$

$$= 18000 - 10000$$

$$= 8000.$$

$$c. \pi_{SR} = TR - TC$$

$$= pQ - (q^2 + 4q + 16)$$

$$= 20(8000) - (8000^2 + 4 \times 8000 + 16)$$

$$= -63,872,016$$

The firms face loss so they would exit the market.

d. In the long run firms earn zero economic profit.

So the $P = MC = AC$

Here
 $c(q, w) = q^r + 4q + 16$

$$MC = \frac{\partial c(q, w)}{\partial q} = 2q + 4$$

$$AC = \frac{c(q, w)}{q} = q + 4 + \frac{16}{q}$$

So, $MC = AC$

$$\Rightarrow 2q + 4 = q + 4 + \frac{16}{q}$$

$$\Rightarrow 2q - q + 4 - 4 - \frac{16}{q} = 0$$

$$\Rightarrow q - \frac{16}{q} = 0$$

$$\Rightarrow q = \frac{16}{q}$$

$$\Rightarrow q^r = 16$$

$$\Rightarrow q^* = 4$$

And $P = MC$

$$\Rightarrow P = 2q + 4$$

$$\Rightarrow P^* = 2 \times 4 + 4 = 12$$

$$\pi_{LR} = Pq - (q^r + 4q + 16) = 12 \times 4 - (4^r + 4 \times 4 + 16) = 0$$

In the long run,

$$c(q, w) = q^r + 4q$$

$$MC = 2q + 4$$

$$AC = q + 4$$

$$MC = AC$$

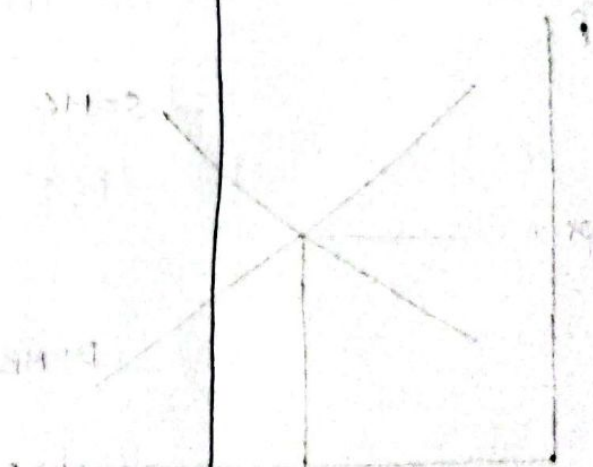
$$2q + 4 = q + 4$$

$$\Rightarrow q^* = 0$$

Also, $P = MC$

$$\Rightarrow P = 2q + 4$$

$$\Rightarrow P^* = 4$$



Welfare and Competitive Markets — Partial Equilibrium

Welfare (increased in \$) is increase of the gains from trade.

Benefit to society minus the cost (measured in money).

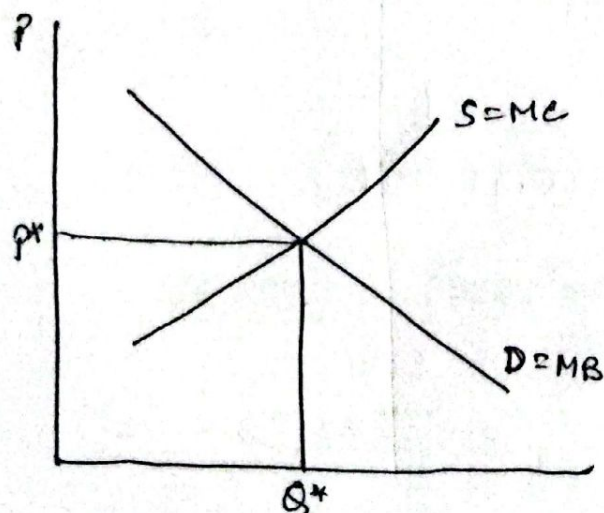
Area under demand and above marginal cost.

(Assuming no externalities nor public goods)

Welfare maximizing output would be, where $MB = MC$.

Demand (as we draw it) = MB

In competitive markets, $S = MC$



Q^* is the quantity that maximizes welfare in that market.

$Q^* \equiv$ socially optimal quantity

where Q^* is the socially optimal quantity

where Q^* is the socially optimal quantity

where Q^* is the socially optimal quantity

where Q^* is the socially optimal quantity

(18)

where Q^* is the socially optimal quantity

(19)

(20)

$$\frac{L}{(p+1)(q+1)} = \frac{m(N)}{1(N)}$$

where Q^* is the socially optimal quantity

General Equilibrium

All markets and the dynamics involved.

Theoretical Approach $\rightarrow$ Welfare Theorems

$\rightarrow$ Existence proofs

$\rightarrow$ Will the law of demands hold

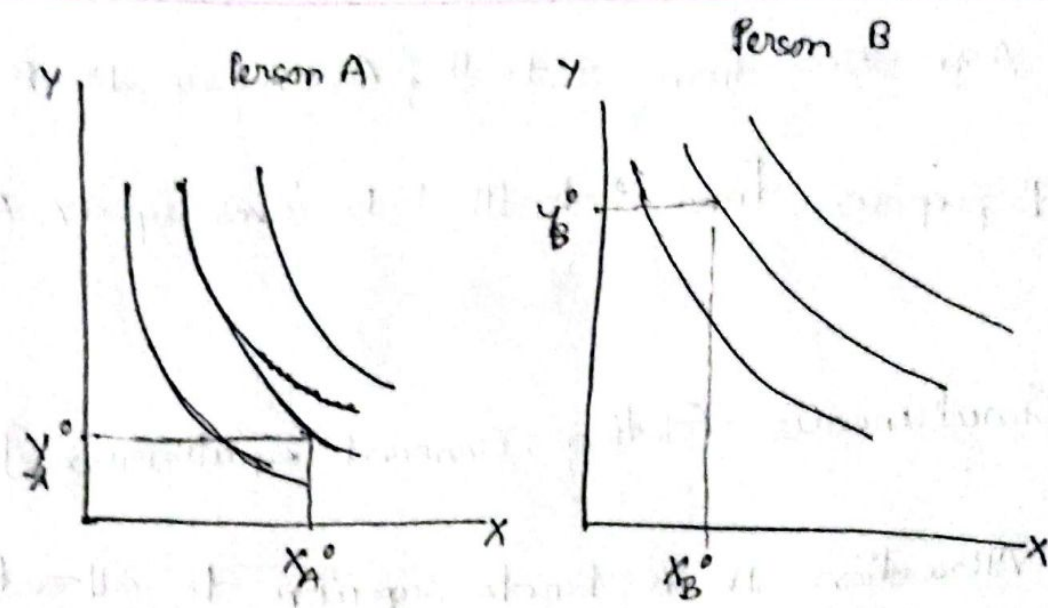
Catch: numbers generated and released are more often than not the gross impacts, rather than the net impact.

Macro dynamics; Stochastic General Equilibrium $\rightarrow$ DSGE

Theory: Exchange Economy

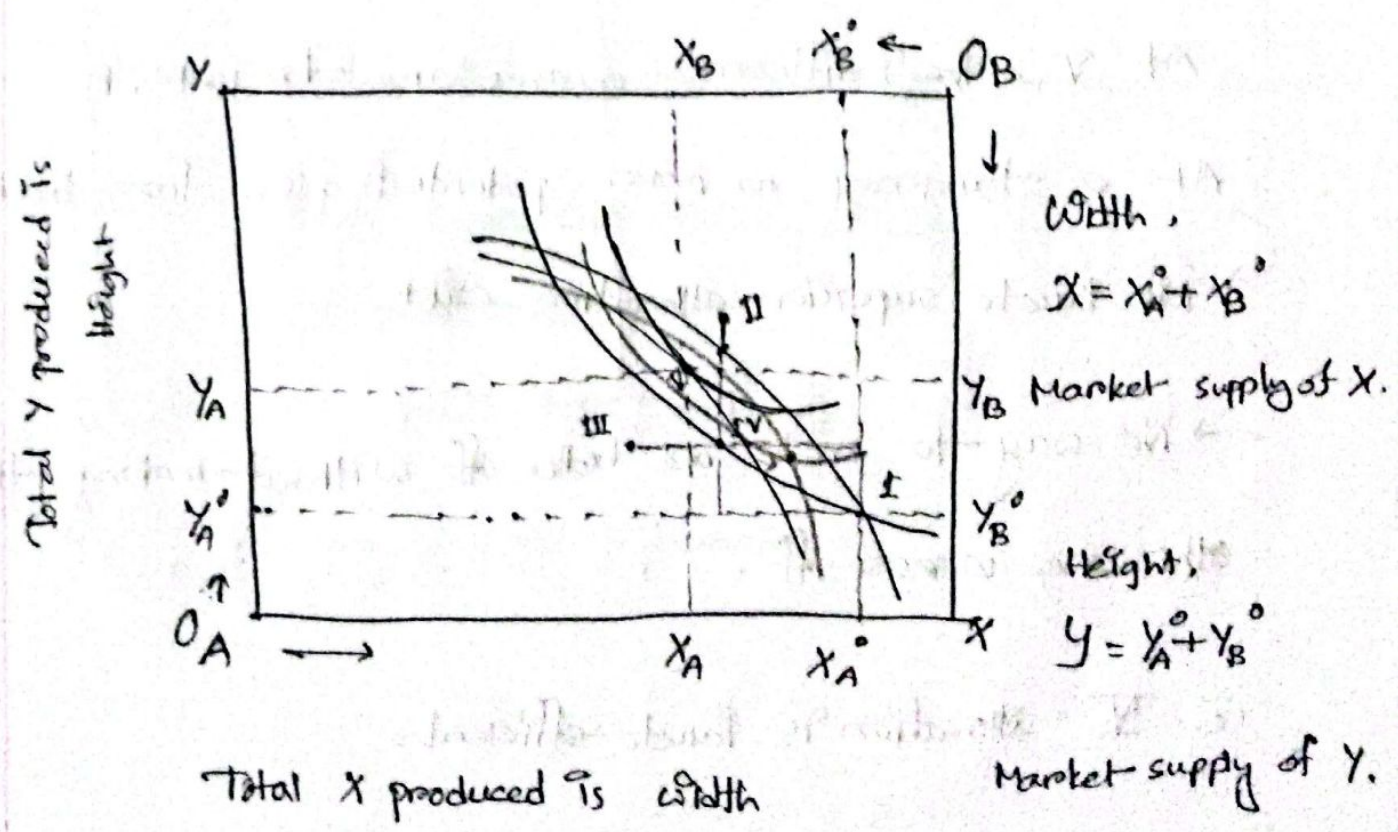
Production has already taken place

Every agent has an endowment which they can freely trade.



Trade happen — When both are mutually benefitted

Core area — the area where both agent are indifferent.



A proposes from I to II : A gives up Δx , B gives up Δy
B proposes from I to III : B gives up Δy , A gives up Δx

[Simultaneous Solution, General equilibrium]

Allocation IV is Pareto superior to allocation I.

Do we stop trading once IV is reached?

After drawing Ics through IV, we see there is still some potential gain from trade.

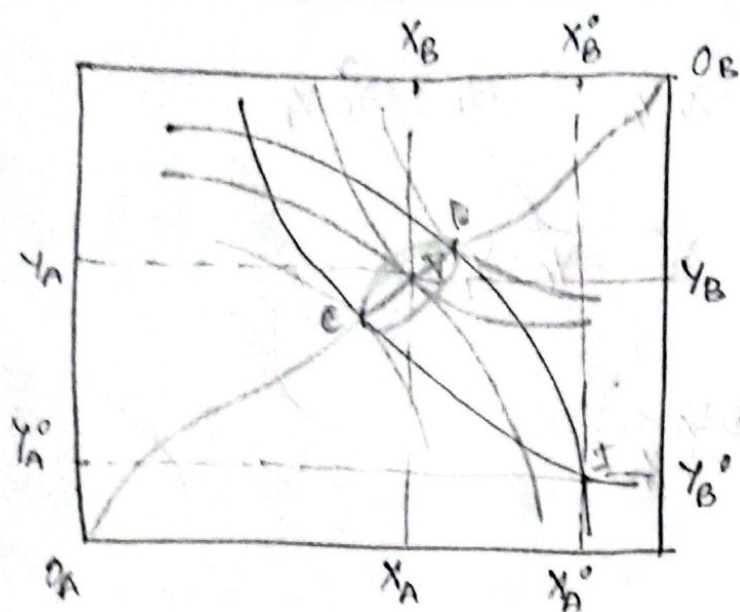
At V, the indifference curves are tangent.

At a tangency, no more potential gains from trade

→ No Pareto superior allocation exists.

→ No way to make one better off without making the other one worse off.

ie V allocation is Pareto-efficient.



The contract curve portion $C_P_{AB} \rightarrow$ core

Core: Part of the contract curve where each party is at least as well off as in the endowment (between C & D)

Q Algebraically find the equation for the contract curve

1. Case 1: Both have diminishing MRS.

2 conditions:

$$\text{Set } MRS_A^A = MRS_B^B$$

$$\text{Supply} = \text{Demand}$$

Example:

$$U_A = X_A Y_A$$

$$U_B = X_B^2 Y_B$$

$$X_A^0 = 50 = X_B^0 = Y_A^0 = Y_B^0$$

$$MRS_A = \left(\frac{\partial U_A}{\partial X_A} \right) / \left(\frac{\partial U_A}{\partial Y_A} \right)$$

$$= \frac{Y_A}{X_A}$$

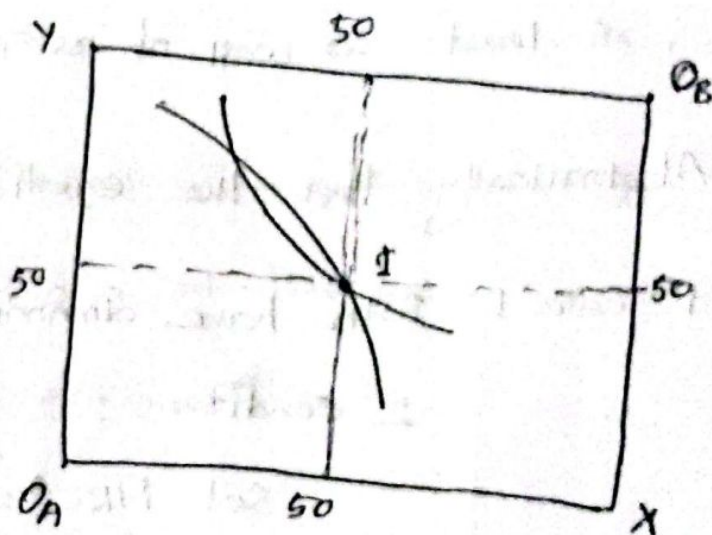
$$MRS_B = \left(\frac{\partial U_B}{\partial X_B} \right) / \left(\frac{\partial U_B}{\partial Y_B} \right)$$

$$= \frac{2X_B Y_B}{X_B^2}$$

$$= \frac{2Y_B}{X_B}$$

$$MRS_A^0 = 1$$

$$MRS_B^0 = 2$$



Since, $MRS_A^0 \neq MRS_B^0$, I is not Pareto-efficient.

The equation for contract curve,

$$\frac{Y_A}{X_A} = \frac{2X_B Y_B}{X_B^2}$$

$$X_A + X_B = 100 = X_A^0 + X_B^0$$

$$Y_A + Y_B = 100 = Y_A^0 + Y_B^0$$

Combining,

$$\frac{Y_A}{X_A} = \frac{2(100 - Y_A)}{100 - X_A} \quad \text{Solve for } Y_A \text{ in terms of } X_A.$$

$$\Rightarrow 100Y_A - X_A Y_A = 200X_A - 2X_A Y_A$$

$$\Rightarrow 100Y_A + X_A Y_A = 200X_A$$

$$\Rightarrow Y_A = \frac{200X_A}{100 + X_A} \quad ; \text{ contract equation}$$

* [If value of X_A , Y_A , X_B & Y_B given, then we can get the answer from graph.]

* [Or get the core]

Case 2: Neither have diminishing MRS (both are linear)

→ the contract curve is the axes (you have to determine which one).

Case 3: One has diminishing MRS, other has constant.

$$U_A = X_A Y_A$$

$$U_B = X_B + 2Y_B$$

$$\rightarrow X_A = 40 \quad X_B = 10 \quad Y_A = 10 \quad Y_B = 40$$

$$MRS_A = \frac{Y_A}{X_A}$$

$$= \frac{10}{40} = \frac{1}{4}$$

$$MRS_A = MRS_B$$

$$MRS_B = \frac{1}{2}$$

$$\Rightarrow \frac{Y_A}{X_A} = \frac{1}{2}$$

$$\frac{Y_A}{X_A} = \frac{1}{2}$$

$$Y_A = \frac{X_A}{2}$$

$$\Rightarrow Y_A = \frac{X_A}{2}$$

→ contract curve.

Contract curve → relation between Y & X

either for Y_A & X_A

or Y_B and X_B .

Case 4: Perfect complements for one, some other utility

with $\frac{dMRS}{dx} \leq 0$

$$U_A = \min(X_A, Y_A)$$

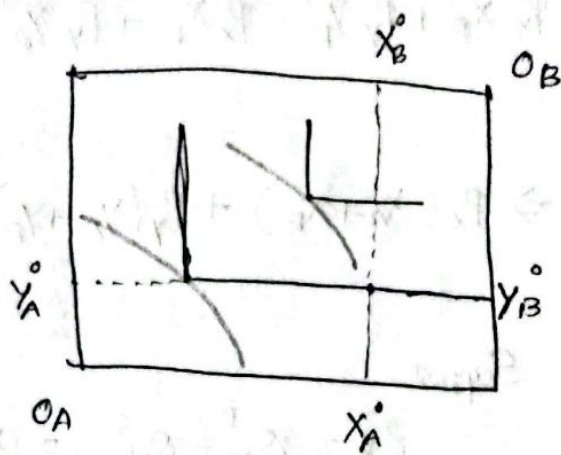
$$U_B = XY$$

$$MRS_A = 0$$

$$MRS_B = \frac{Y}{X}$$

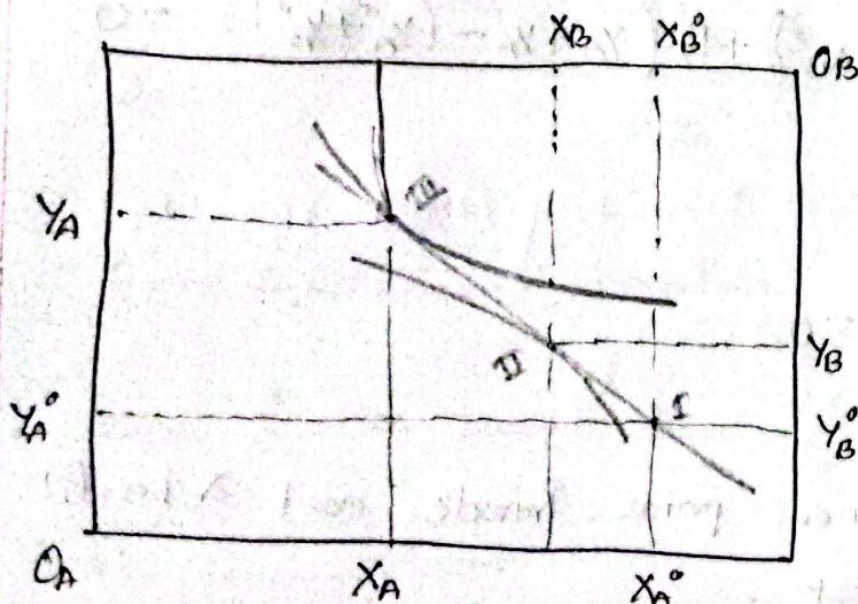
contract curve, $X_A = Y_A$

$$\frac{Y}{X} = 0 \Rightarrow Y = 0$$



Budget line equation $P_X X_A^0 + P_Y Y_A^0 = I_A$

Slope = $-\frac{P_X}{P_Y}$ and goes through the endowment. I.



$$Y_A + Y_B > Y_A^o + Y_B^o \rightarrow Q_Y^D > Q_Y^S$$

$$X_A + X_B < X_A^o + X_B^o \rightarrow Q_X^D < Q_X^S$$

$$P_X X_A + P_Y Y_A = P_X X_A^o + P_Y Y_A^o$$

$$P_X X_B + P_Y Y_B = P_X X_B^o + P_Y Y_B^o$$

$$\Rightarrow P_X (X_A + X_B) + P_Y (Y_A + Y_B) = P_X (X_A^o + X_B^o) + P_Y (Y_A^o + Y_B^o)$$

Suppose

$$Z_X = Q_X^D - Q_X^S = X_A + X_B - (X_A^o + X_B^o)$$

$$Z_Y = Y_A + Y_B - (Y_A^o + Y_B^o)$$

$$P_X Z_X + P_Y Z_Y = 0$$

$$\Rightarrow P_X (X_A + X_B - (X_A^o + X_B^o)) + P_Y (Y_A + Y_B - (Y_A^o + Y_B^o)) = 0$$

Walras' law,

$$\sum_{i=1}^n P_i Z_i(P) \equiv 0$$

No one can influence price. Private cost & benefit equals social benefit & cost.

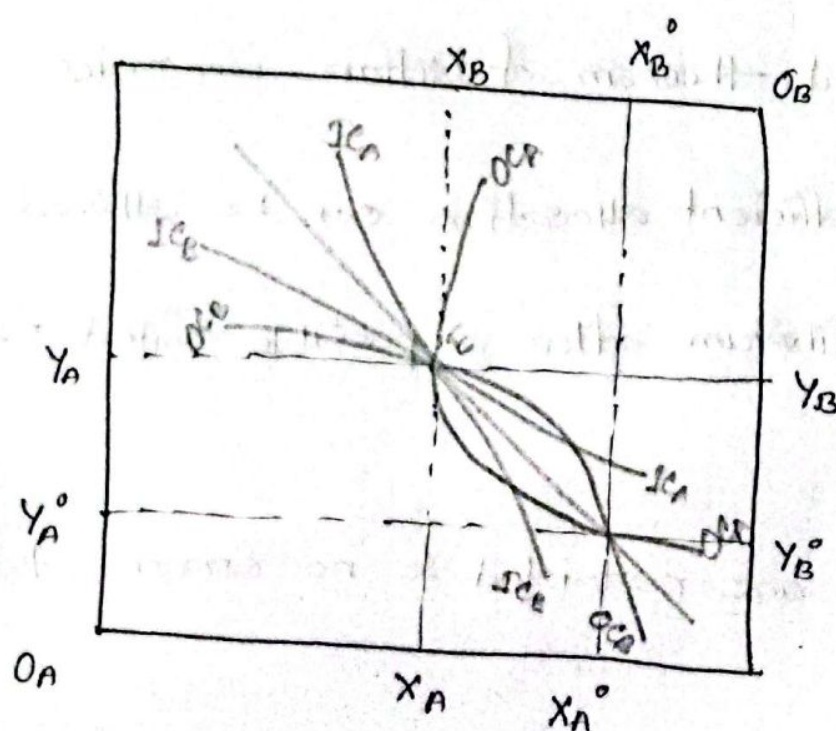
Social benefit $\rightarrow$ Demand Social cost $\rightarrow$ Supply ?

Walrasian equilibrium — Pareto-efficient

A market is in equilibrium when $z_i(p) = 0$

then the economy is in a general equilibrium

when $z_i(p^*) = 0, \forall i$



$E \rightarrow$ Equilibrium allocation

Slope of the budget line $\frac{P_x^*}{P_y^*}$

Note: Pareto-efficient.

1st - fundamental Theorem of Welfare Economics:

If the allocation results from a Walrasian equilibrium, it will be Pareto-efficient.

Walrasian Markets have 2 properties

- No one can influence the price (which easily adjust)
- Private cost/benefit = social cost/benefits.

2nd fundamental theorem of welfare economics:

Any Pareto-efficient allocation can be attained by a Walrasian equilibrium after a possible initial reallocation of resources.

(Preferences are restricted so no corner solutions)

Homework — 20th February (most probably)

Tutorial — 27th February.